David Xchel Morales Hurtado

Systems Engineer — Hardware-to-Cloud | Distributed Systems | Staff Engineer Track

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Professional Summary

Systems Engineer with 5+ years professional Python/Bash automation (Intel, Oracle) and strong C++ foundation from academic, personal projects (MultiApp C++20), and open-source contributions (Blender, Godot, fmt, CXXGraph). Bridging hardware-software boundary for performance and reliability. Actively learning distributed systems (DDIA, system design). Targeting **Staff Engineer** track in distributed systems and platform engineering.

Selected Achievements

- **Bronze Medalist** - Mexican Mathematics Olympiad (OMM).
- **Reduced validation execution time** from **hours to minutes** by building **Python/Bash automation frameworks**.
- **Improved testing efficiency by 50%** across multiple processor validation programs.
- **Served as technical mentor and onboarding lead** for incoming engineers.
- **Active open-source contributor**:
  - Blender: Fixed a custom property type bug.
  - fmt: PR for resolving an Format overflow bug.
  - Godot Engine: PR for removing a duplicate-buffer deletion.
  - CXXGraph: Added a large graph cycle detection test.
  - UnifyAI: Extended the test base with dot product PyTest cases.
- Completed an **international academic exchange** at **Universiti Malaya**.

Professional Experience

Oracle — Guadalajara, Mexico

Feb 2026 – Present

Systems Engineer — Cloud Infrastructure

Develop cloud-management tooling in **Python**, **Java**, and **Bash** for Oracle Cloud Infrastructure regions.

- Build and maintain automation workflows for cloud-region deployment, reducing manual effort and improving **reliability**.
- Troubleshoot **Linux** systems and **cloud** infrastructure across production environments.
- **Develop automation tooling** and AI-assisted tooling for information extraction and operational support.
- **Authored** technical workflows and onboarding material adopted across engineering teams.

Intel — Guadalajara, Mexico

Feb 2021 – Feb 2026

Systems Engineer — Validation & Automation

Develop **Python** and **Bash** automation for functional validation of next-generation processor architectures on **Linux**, collaborating with a globally distributed team.

- Designed and maintained large-scale validation automation frameworks using **Python**, **Bash**, and **PyTest**.
- **Reduced execution time** of manual workflows from hours to minutes through **automation**.
- **Built** migration tooling that reduced project-porting effort **from one day to minutes**.
- **Developed CI/CD** validation pipelines supporting multiple processor programs.
- Acted as **technical lead** and primary **point of contact** across cross-functional teams.
- **Mentored** new engineers on processor architecture, memory systems, and **Linux**-based development workflows.
- Improved validation **efficiency by 50%** through **optimized** test coverage and execution strategy.
- **Applied control systems theory** to distributed validation pipelines; designed fault-injection frameworks mapping to chaos engineering principles.

Education

Instituto Politécnico Nacional — Mexico City

Aug 2015 – Jan 2021

B.Eng. in Mechatronics Engineering

Top-performing student recognized for **effort**, **curiosity**, and **creativity**. Consistently focused on **teamwork** by **mediating** and **leading** to deliver quality projects efficiently.

- **MIPS** processor implementation in **SystemVerilog** with a **UVM testbench** built using **Python** and **C++**.
- **Python** projects: **Neural Networks**, **file/database processing**, **web scraping**, **Flask** applications with Selenium testing.
- **Virtual reality** experiences developed in **Unity** and **Blender**.
- Embedded systems and FPGA development projects, electronic circuit design and implementation.

Universiti Malaya — Kuala Lumpur, Malaysia
Feb 2019 – Jun 2019

Academic Mobility / Exchange Program

Learned from one of the most renowned universities in the world while acquiring experience by working, **communicating** and **leading** people with contrasting backgrounds and mindsets.

- Built a line-following robot using **Raspberry Pi**.
- Interactive **VR** world using **Unity**.
- Developed a hand-tracking **AI** project.

Skills

Languages	**Python**, **Bash**, **Java**, **C++20**, C, Verilog
Platforms	**Linux**, **Docker**, Git/GitHub, Kubernetes (basic)
Engineering	**Distributed Systems (self-study)**, **System Design (self-study)**, **Observability**, CI/CD, Automation, Embedded Systems
Professional	**Leadership**, **Mentoring**, **Technical Communication**, **Problem Solving**
Spoken Languages	**English (fluent)**, **Spanish (native)**, German (learning), French (learning)

Personal Projects

- **MultiApp** — Modular C++20 desktop application (GTK4, WebKitGTK, asio, GTest) with plugin architecture, embedded browser, multithreaded fractal viewer, coroutine-based LAN chat. github.com/dxchel/MultiApp
- **Personal Portfolio** — dxchel.github.io, React 19 + TypeScript + Bootstrap 5, Create React App, GitHub Pages.
- **Open Source Contributions** — Blender, Godot Engine, fmt, CXXGraph, UnifyAI.
- **Personal AI Agent** — Local AI agent for research, data organization, workflow automation.
- **Arch Linux Daily Driver** — Customized system for deep Linux internals knowledge.

Courses & Certifications

Advanced C++ Programming and Modern Practices — Microsoft	Feb 2026 – May 2026
Linux Essentials — Linux Professional Institute	Jan 2020 – Apr 2020
Advanced Python — Universidad Marista	Jun 2024 – Nov 2024
Time Management Fundamentals	Feb 2026 – Mar 2026
Foundations of Agile Project Management — Google	Feb 2026 – Mar 2026
Learning How to Learn — Oakland University	Jan 2025 – Jun 2025
Mobile App Development — Google Activate	Nov 2019 – Feb 2020
Intel x86 Processor & Platform Architecture — MindShare	Jan 2022 – Jun 2022

Extra-Curricular Activities

Mexican Mathematics Olympiad (OMM), Mexico City — Mar 2013 – Nov 2014
— Bronze award

Mexican Informatics Olympiad (COMI), Mexico City — Jan – May 2014
— Participation award

Robot War Staff, IEEE University Branch, Mexico City — Jul 2018 – Jul 2020
— Planning and organization staff

Scout, Mexican Scout Association, Morelia — Jan 2014 – Aug 2015
— Participation in altruism and teamwork-oriented activities

Emergency Response Helper, ERUM, Mexico — Sep – Oct 2017
— Provided relief support following the 2017 Mexico City earthquake